OS lab CPU SCHEDULING ALGORITHMS

## 1.FIRST COME FIRST SERVE (FCFS)

CODE:

```
#include <stdio.h>
int main(){
   int n;
   printf("Enter number of processes: ");
   scanf("%d",&n);
   int pid[n], at[n], bt[n], ct[n], tat[n], wt[n], rt[n];
   for(int i=0;i<n;i++){
      pid[i] = i+1;
      printf("Enter AT and BT for P%d: ",pid[i]);
      scanf("%d %d",&at[i],&bt[i]);
   }
   for(int i=0;i<n-1;i++){
      for(int j=i+1;j<n;j++){
         if(at[i] > at[j]){
            int temp;
            temp = at[i];
            at[i] = at[j];
            at[j] = temp;
            temp = bt[i];
            bt[i] = bt[j];
            bt[j] = temp;
            temp = pid[i];
            pid[i] = pid[j];
            pid[j] = temp;
         }}}
   int time = 0;
   for(int i=0;i<n;i++){
      if(time < at[i])
         time = at[i];
      rt[i] = time - at[i];
      time = time + bt[i];
      ct[i] = time;
   }
```

```
    float totalTAT=0,totalWT=0,totalRT=0;
    for(int i=0;i<n;i++){
        tat[i] = ct[i] - at[i];
        wt[i] = tat[i] - bt[i];
        totalTAT += tat[i];
        totalWT += wt[i];
        totalRT += rt[i]; }
    printf("\nPID\tAT\tBT\tCT\tTAT\tWT\tRT\n");
    for(int i=0;i<n;i++){
        printf("P%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
        pid[i],at[i],bt[i],ct[i],tat[i],wt[i],rt[i]);
    }
    printf("\nAverage TAT = %.2f",totalTAT/n);
    printf("\nAverage WT  = %.2f",totalWT/n);
    printf("\nAverage RT  = %.2f\n",totalRT/n);
    return 0;
}
```

Out put:

```
Enter number of processes: 4
Enter AT and BT for P1: 0 7
Enter AT and BT for P2: 2 4
Enter AT and BT for P3: 4 1
Enter AT and BT for P4: 5 4

PID     AT      BT      CT      TAT     WT      RT
P1      0       7       7       7       0       0
P2      2       4       11      9       5       5
P3      4       1       12      8       7       7
P4      5       4       16      11      7       7

Average TAT = 8.75
Average WT  = 4.75
Average RT  = 4.75
PS D:\vishwanath\OS LAB>
```

2.SHORTEST JOB FIRST (SJF) NON preemtive

Code:

```
#include <stdio.h>
int main(){
   int n,i;
   int at[10],bt[10],wt[10],tat[10],ct[10],rt[10];
   int completed[10]={0};
   int time=0,count=0,min,index;
   float avg_wt=0,avg_tat=0;
   printf("Enter number of processes: ");
   scanf("%d",&n);
   for(i=0;i<n;i++)  {
      printf("Enter AT and BT for P%d: ",i+1);
      scanf("%d%d",&at[i],&bt[i]);
   }
   while(count<n){
      min=9999;
      index=-1;
      for(i=0;i<n;i++){
         if(at[i]<=time && completed[i]==0 && bt[i]<min) {
            min=bt[i];
            index=i;
         }}
      if(index==-1){
         time++;
      }
      else {
         wt[index]=time-at[index];
         rt[index]=wt[index];
         time=time+bt[index];
         ct[index]=time;
         tat[index]=ct[index]-at[index];
         completed[index]=1;
         count++;
         avg_wt+=wt[index];
         avg_tat+=tat[index];
```

```
        }
    }

    printf("\nP\tAT\tBT\tCT\tWT\tTAT\tRT\n");

    for(i=0;i<n;i++)
    {
        printf("P%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
        i+1,at[i],bt[i],ct[i],wt[i],tat[i],rt[i]);
    }

    printf("\nAverage Waiting Time = %.2f",avg_wt/n);
    printf("\nAverage Turnaround Time = %.2f\n",avg_tat/n);

    return 0;
}
```

Output:

```
Enter number of processes: 4
Enter AT and BT for P1: 0 7
Enter AT and BT for P2: 8 3
Enter AT and BT for P3: 3 4
Enter AT and BT for P4: 5 6

P       AT      BT      CT      WT      TAT     RT
P1      0       7       7       0       7       0
P2      8       3       14      3       6       3
P3      3       4       11      4       8       4
P4      5       6       20      9       15      9

Average Waiting Time = 4.00
Average Turnaround Time = 9.00
PS D:\vishwanath\OS LAB>
```

3.SHORTEST REMAINING TIME FIRST (SRTF) -SJF PREEMTIVE

Code:

```
#include <stdio.h>
int main(){
   int n,i,time=0,completed=0;
   int at[10],bt[10],rbt[10];
   int ct[10],wt[10],tat[10],rt[10],start[10];
   float avg_wt=0,avg_tat=0,avg_rt=0;
   printf("Enter number of processes: ");
   scanf("%d",&n);
   for(i=0;i<n;i++){
      printf("Enter AT and BT for P%d: ",i+1);
      scanf("%d%d",&at[i],&bt[i]);
      rbt[i]=bt[i];
      start[i]=-1;
   }
   while(completed<n){
      int smallest=-1;
      for(i=0;i<n;i++){
         if(at[i] <= time && rbt[i] > 0){
            if(smallest == -1 ||
               rbt[i] < rbt[smallest] ||
               (rbt[i] == rbt[smallest] && at[i] < at[smallest])){
                smallest = i;
            }} }
      if(smallest == -1){
         time++;
         continue;
      }
      if(start[smallest] == -1)
         start[smallest] = time;
      rbt[smallest]--;
      time++;
      if(rbt[smallest] == 0){
         ct[smallest] = time;
         tat[smallest] = ct[smallest] - at[smallest];
```

```
        wt[smallest] = tat[smallest] - bt[smallest];
        rt[smallest] = start[smallest] - at[smallest];
        avg_wt += wt[smallest];
        avg_tat += tat[smallest];
        avg_rt += rt[smallest];
        completed++;
      }
    }
    printf("\nP\tAT\tBT\tCT\tWT\tTAT\tRT\n");
    for(i=0;i<n;i++){
      printf("P%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
      i+1,at[i],bt[i],ct[i],wt[i],tat[i],rt[i]);
    }
    printf("\nAverage WT = %.2f",avg_wt/n);
    printf("\nAverage TAT = %.2f",avg_tat/n);
    printf("\nAverage RT = %.2f\n",avg_rt/n);

    return 0;
}
```

Output:

```
Enter number of processes: 4
Enter AT and BT for P1: 0 7
Enter AT and BT for P2: 8 3
Enter AT and BT for P3: 3 4
Enter AT and BT for P4: 5 6

P       AT      BT      CT      WT      TAT     RT
P1      0       7       7       0       7       0
P2      8       3       14      3       6       3
P3      3       4       11      4       8       4
P4      5       6       20      9       15      9

Average WT = 4.00
Average TAT = 9.00
Average RT = 4.00
PS D:\vishwanath\OS LAB>
```

4.PRIORITY NON PREEMTIVE

Code:

```
#include <stdio.h>
int main(){
   int n,i,time=0,count=0,highest;
   int at[10],bt[10],pr[10];
   int ct[10],wt[10],tat[10],rt[10];
   int completed[10]={0};
   float avg_wt=0,avg_tat=0,avg_rt=0;
   printf("Enter number of processes: ");
   scanf("%d",&n);
   for(i=0;i<n;i++){
      printf("Enter AT, BT and Priority for P%d: ",i+1);
      scanf("%d%d%d",&at[i],&bt[i],&pr[i]);
   }
   while(count<n){
      highest=-1;
      for(i=0;i<n;i++){
         if(at[i]<=time && completed[i]==0){
            if(highest==-1 || pr[i] < pr[highest]){
               highest=i;
            }
         }
      }
      if(highest==-1){
         time++;
         continue;
      }
      rt[highest]=time-at[highest];
      time += bt[highest];
      ct[highest]=time;
      tat[highest]=ct[highest]-at[highest];
      wt[highest]=tat[highest]-bt[highest];
      avg_wt += wt[highest];
      avg_tat += tat[highest];
      avg_rt += rt[highest];
```

```
        completed[highest]=1;
        count++;
    }
    printf("\nP\tAT\tBT\tPR\tCT\tWT\tTAT\tRT\n");
    for(i=0;i<n;i++){
        printf("P%d\t%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
        i+1,at[i],bt[i],pr[i],ct[i],wt[i],tat[i],rt[i]);
    }
    printf("\nAverage WT = %.2f",avg_wt/n);
    printf("\nAverage TAT = %.2f",avg_tat/n);
    printf("\nAverage RT = %.2f\n",avg_rt/n);
    return 0;
}
```

Output:

```
Enter number of processes: 5
Enter AT, BT and Priority for P1: 0 3 5
Enter AT, BT and Priority for P2: 2 2 3
Enter AT, BT and Priority for P3: 3 5 2
Enter AT, BT and Priority for P4: 4 4 4
Enter AT, BT and Priority for P5: 6 1 1

P       AT      BT      PR      CT      WT      TAT     RT
P1      0       3       5       3       0       3       0
P2      2       2       3       11      7       9       7
P3      3       5       2       8       0       5       0
P4      4       4       4       15      7       11      7
P5      6       1       1       9       2       3       2

Average WT = 3.20
Average TAT = 6.20
Average RT = 3.20
PS D:\vishwanath\OS LAB>
```

5.PRIORITY PREEMTIVE

Code:

```
#include <stdio.h>
struct Process{
   int pid, at, bt, pr, rt, ct, wt, tat, start, response;
};
int main(){
   struct Process p[10];
   int n,i,time=0,completed=0,highest;
   float avg_wt=0,avg_tat=0,avg_rt=0;
   printf("Enter number of processes: ");
   scanf("%d",&n);
   for(i=0;i<n;i++){
      printf("Enter AT, BT and Priority for P%d: ",i+1);
      scanf("%d%d%d",&p[i].at,&p[i].bt,&p[i].pr);
      p[i].pid=i+1;
      p[i].rt=p[i].bt;
      p[i].start=-1;
   }
   while(completed<n){
      highest=-1;
      int min_pr=9999;
      for(i=0;i<n;i++){
         if(p[i].at<=time && p[i].rt>0 && p[i].pr<min_pr){
            min_pr=p[i].pr;
            highest=i;
          ] }
      if(highest==-1){
         time++;
      }
      else{
         if(p[highest].start==-1){
            p[highest].start=time;
            p[highest].response=p[highest].start-p[highest].at;
            avg_rt+=p[highest].response;
         }
```

```
        p[highest].rt--;
        time++;
        if(p[highest].rt==0){
          completed++;
          p[highest].ct=time;
          p[highest].tat=p[highest].ct-p[highest].at;
          p[highest].wt=p[highest].tat-p[highest].bt;
          avg_wt+=p[highest].wt;
          avg_tat+=p[highest].tat;  } }   }
  printf("\nPID\tAT\tBT\tPR\tCT\tWT\tTAT\tRT\n");
  for(i=0;i<n;i++){
    printf("P%d\t%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
    p[i].pid,p[i].at,p[i].bt,p[i].pr,
    p[i].ct,p[i].wt,p[i].tat,p[i].response);
  }
  printf("\nAverage Waiting Time = %.2f",avg_wt/n);
  printf("\nAverage Turnaround Time = %.2f",avg_tat/n);
  printf("\nAverage Response Time = %.2f\n",avg_rt/n);
  return 0;
}
```

Output:

```
Enter number of processes: 5
Enter AT, BT and Priority for P1: 0 3 5
Enter AT, BT and Priority for P2: 2 2 3
Enter AT, BT and Priority for P3: 3 5 2
Enter AT, BT and Priority for P4: 4 4 4
Enter AT, BT and Priority for P5: 6 1 1

PID     AT      BT      PR      CT      WT      TAT     RT
P1      0       3       5       15      12      15      0
P2      2       2       3       10      6       8       0
P3      3       5       2       9       1       6       0
P4      4       4       4       14      6       10      6
P5      6       1       1       7       0       1       0

Average Waiting Time = 5.00
Average Turnaround Time = 8.00
Average Response Time = 1.20
PS D:\vishwanath\OS LAB>
```